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Image encryption using memristive hyperchaos

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Abstract
In this paper, a novel image encryption algorithm (DMHM-IEA) based on a two-dimensional discrete memristive hyperchaotic map (2D-DMHM) derived from discrete memristor and improved Logistic map is proposed. The thorough performance investigation demonstrates that adding the memristor enhances the complicated dynamic characteristics and sequence unpredictability of the proposed map, making it more appropriate for usage in image encryption applications. Diffusion-permutation-diffusion is an inventive structure that DMHM-IEA uses to provide great encryption effect with just one round of encryption. In the random position permutation, pixel positions are quickly and arbitrarily changed to new positions, while the diffusion process uses main-diagonal diffusion and counter-diagonal diffusion to spread out the minor changes in the original image throughout the entire plane, resulting in a highly distinct encrypted image. The combined performance and security evaluation demonstrates that DMHM-IEA has strong tolerance to noise and data loss as well as good resistance to potential attacks. When compared to other cutting-edge algorithms, the proposed algorithm performs well in terms of security.

Keywords Chaos · Discrete memristor · Hyperchaotic map · Image encryption · Performance analysis

1 Introduction

With the exponential development of the Internet and digital technology, the digital image is rapidly spreading across the web as the most visual and common type of data. The majority of digital images sent across networks are private, such as national military photographs, private company images, and individual medical pictures, which cannot be accessed by unauthorized third parties. It is possible to illegally capture and decrypt transmitted images thanks to the advancement of network technology, which could result in the illegal publication and fabrication of personal data [1]. Therefore, it is absolutely crucial to secure the components of these images. Image encryption is the most simple and effective technique among these security methods, and it may transform the private images into unidentifiable noise-like images for transmission and significantly prevent the revelation of the sensitive information.

The classic encryption methods DES, AES and RSA, which are mostly used for text data encryption, are thought to be capable of encrypting images, but because of improving image quality, they must compromise time efficiency in order to maintain algorithm security [2]. The most appealing of the many image encryption algorithms using various techniques is chaos cryptography because the randomness, ergodicity, determinism, and initial condition sensitivity of chaos coincide with many requirements of encryption and can be perfectly related to the concepts of confusion and diffusion in conventional cryptographic theory [3]. Permutation and diffusion are the most frequent approaches in image encryption algorithms [4], where permutation can be achieved through changing pixel positions and diffusion can be accomplished by affecting the entire image pixels with tiny changes. In the whole process, the noise-like encrypted image can be converted into the original image only by using the correct security key, otherwise the decrypted image will still be a noise-like image. Depending on the chaotic sequence generation used by the reported image encryption algorithms, there are usually two different approaches: (i) continuous chaotic systems and (ii) discrete chaotic maps. According to a perceptron model in a neural network built on a high-dimensional Lorenz chaotic system, Wang et al. [5] suggested a chaotic image encryption method. Wang et al. [6] reported an image encryption algorithm based on a four-dimensional hidden attractor hyperchaotic system and Knuth-Durstenfeld algo-

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rithm. The chaos-generated sequence is crucial to the encryption algorithm. The level of randomness in the sequence influences whether the encryption effect of the algorithm can be maximized, and the time it takes to generate the sequence has an impact on how long the image encryption procedure takes to complete as a whole. The generation of chaotic sequences for image encryption requires discretization of continuous systems, however discrete maps remove this step and speed up the generation of chaotic sequences [7]. Using coupled Logistic map and cubic map based on chunked compressive sensing and multiple S-boxes, Nan et al. [8] suggested a remote sensing image compression and encryption algorithm. By coupling the sinusoidal discrete memristor with the Gaussian map, Lai et al. [9] devised a plaintext correlated image encryption algorithm that can be implemented in a single round and uses the chaotic matrix and the position information of the original image to achieve the position exchange of pixels.

Some other researchers have studied the pixel in more depth [10, 11]. In their new method for bit-level replacement, Raza et al. [12] use 3D puzzles built on chaotic map. Based on a 5D multi-wing hyperchaotic system, Li et al. [13] presented an image encryption algorithm employing pixel-level permutation and bit-level permutation. The pixel-level permutation process entirely obscures the original image by destroying the correlation between neighboring pixels, while the bit-level permutation process directly modifies the bits of the pixels by combining four bytes to modify the bits of each byte.

Aside from these methods, others are thought to be beneficial in enhancing the security of the encryption algorithm, including integrating different techniques [14, 15], expanding the key space [16], improving the structure of the encryption algorithm [17], and boosting the unpredictability of the sequence [18]. Feng et al. [19] suggested an image encryption algorithm based on five-dimensional hyperchaotic map in combination with DNA encoding. Zhang et al. [20] proposed a 3D Zigzag transform based on two 1D chaotic systems inspired by the three-view drawing in the engineering field. Using 2D-LACM paired with Logistic map and Sine map, Zheng et al. [21] suggested a dynamic S-box based image encryption algorithm. To address the shortcomings of the algorithm in terms of security and efficiency, Chai et al. [22] proposed a color image encryption algorithm that combines hybrid multi-objective particle swarm optimization algorithm, chunked compression-aware technique, and Hessenberg decomposition approach. Gao et al. [23] constructed a key space that is substantially higher than the required minimum to resist brute-force attacks. Ye et al. [24] presented an permutation-rewriting-diffusion-based pixel-level image encryption algorithm that can improve the relationship between pixel position mixing and gray value changes. Prior to permutation and diffusion, Lai et al. [25] suggested a pixel-split image encryption algorithm based on a 2D Salomon map to exchange the high and low bits of the image pixels randomly. Based on the one-dimensional discrete chaotic map obtained by coupling Tent map and Sine map, Zhou et al. [26] proposed an image encryption scheme with excellent confusion and diffusion properties. Wu et al. [27] created a new image encryption algorithm based on coupling Hénon map and Sine map to generate the 2D-HSM with improved ergodicity and pseudo-randomness to encrypt the original image utilizing chaotic sequences and DNA operating rules, however regrettably the map lacks a more sophisticated hyperchaotic behavior. Ye et al. [28] designed a new image encryption algorithm based on the chaotic key generated by the memristive chaotic circuit system. Lai et al. [29] constructed a novel memristor Hopfield neural network and designed a 3D image encryption scheme using the generated multiple double vortex attractors.

There are reports of increasingly chaotic image encryption algorithms to protect image data, however as the computing power of computers increases, some of these algorithms are vulnerable to attack, particularly when using plaintext attacks [30]. Chen et al. [31] show that the Hua et al. suggested high-speed permutation and pixel adaptive diffusion medical image encryption technique is capable of properly recovering encrypted images via chosen-plaintext attacks under bad randomness. Liu et al. [32] examined the image encryption algorithm created by Behzad et al. based on 1D-CCSM and the dynamic permutation method of block size, and they demonstrated that the algorithm cannot be utilized for secure communication since it is not immune to chosen-plaintext attacks.

In the meantime, some image encryption algorithms choose to increase the quantity of permutation and diffusion rounds in order to maintain the encryption effect. The robustness of the algorithm to noise interference and cropping attacks declines as the number of rounds increases, making this method susceptible to signal interference and other transmission-related factors even though it can achieve a good encryption effect [33]. Several image encryption algorithms in the aforementioned literature require at least two rounds to have a reasonable encryption effect, and they all share the problem of having low resistance to outside interference [34].

Consequently, it is crucial to develop an image encryption algorithm based on the chaotic sequence with high pseudo-randomness and complexity that is also robust to noise and data loss and resistant to potential attacks such chosen-plaintext attacks.

1.1 Contributions

Aiming at the problems of the existing image encryption algorithms, such as vulnerability to potential attacks like

the chosen-plaintext attack, weakened robustness due to the excessive number of encryption rounds, and the vulnerability of the encryption algorithm structure itself, an image encryption algorithm with a diffusion-permutation-diffusion structure is designed based on an discrete memristive hyperchaotic map consisting of a memristor coupled with an improved Logistic map. The main contributions and innovations of this study are summarized below.

(1) The 2D-DMHM produced by combining discrete memristor with improved Logistic map significantly improves its sensitivity to initial values while eliminating the drawbacks of Logistic, such as simple behavior and fragile chaotic intervals.
(2) The performance is analyzed in terms of Lyapunov exponents, bifurcation diagrams and chaotic trajectory. The 2D-DMHM has continuous and wide chaotic range, strong sequence randomness, and the addition of the memristor causes it to exhibit hyperchaotic behavior.
(3) Based on the 2D-DMHM, a diffusion-permutation-diffusion encryption structure is created, and a secure and effective image encryption algorithm is proposed using random position permutation, main-diagonal diffusion, and counter-diagonal diffusion.
(4) With only one round of encryption, the proposed algorithm has excellent confusion and diffusion properties and is highly resistant to a variety of potential attacks.
(5) As compared to existing image encryption algorithms, simulation evaluation and comparison demonstrate that DMHM-IEA has more stable encryption state, greater security level, and higher encryption efficiency.

The rest of this paper is organized as follows. Section 2 gives a brief description of the preliminary work, mainly illustrating the constructed discrete memristive hyperchaotic map, Section 3 explains the specific procedure of the proposed DMHM-IEA permutation and diffusion, Section 4 provides a comprehensive security performance analysis of the proposed algorithm, and the whole paper is finally concluded in Section 5.

2 Preliminaries

This section introduces the properties of the memristor, the improved logistic map, and the novel proposed discrete memristor hyperchaotic map. It also discusses the complexity of the proposed map by examining the Lyapunov exponent, chaotic trajectory, bifurcation diagram, and randomness test of the sequences.

2.1 Discrete memristor

Suppose $q(t)$ is the internal charge variable, $M(q) = sinq(t)$ is expressed as the periodic memristor function, and $i(t)$ and $v(t)$ are indicated as the input current and output voltage, the general ideal charge control memristor can be described as

$$\begin{aligned} v(t) &= M(q) \cdot i(t) = \sin q(t) \cdot i(t) \\ dq(t)/dt &= i(t) \end{aligned} \tag{1}$$

According to the discrete method of continuous-time memristor by [35], assume that the current $i(t)$, voltage $v(t)$, and charge $q(t)$ at the nth iteration are denoted as i_n, v_n, and q_n, respectively, the mathematical model of the discrete memristor can be expressed as

$$\begin{aligned} v_n &= M(q_n) \cdot i_n = \sin q_n \cdot i_n \\ q_{n+1} &= q_n + h \cdot i_n \end{aligned} \tag{2}$$

where q_{n+1} is the sampled value of $q(t)$ at the $(n+1)$-th iteration, the constant h is typically 1. A discrete sinusoidal current $i_n = Asin(\omega n)$ is chosen as the excitation in order to further confirm the fundamental characteristics of the discrete memristor model. Figure 1 illustrates the pinch hysteresis loops of the discrete memristor with frequency and amplitude dependent variation. The outcomes demonstrate that the memristor can capture the memory effect and satisfy all of its fundamental properties.

2.2 Improved Logistic map

The choice of initial conditions for each chaotic map is shown in Table 1, and it can be seen that the Logistic only includes one parameter and one variable, giving the researcher more possibilities for improvement. Overall, Logistic map is a powerful tool for studying chaotic dynamics and has numerous applications in various fields. Its simplicity and ability to capture complex behaviors make it a valuable tool for researchers and practitioners alike. The Logistic map and the classic Sine map are combined to create an improved Logistic map, which is described as

$$x_{n+1} = \mu x_n \sin(a(1 - x_n)) \tag{3}$$

where the control parameter $\mu \in [0, 40]$. Figure 3(a)-(b) show the bifurcation diagram and Lyapunov exponent of the original Logistic map and the improved Logistic map, respectively. It can be found from the figure that the chaotic range of the improved Logistic map is much larger than that of the original map.

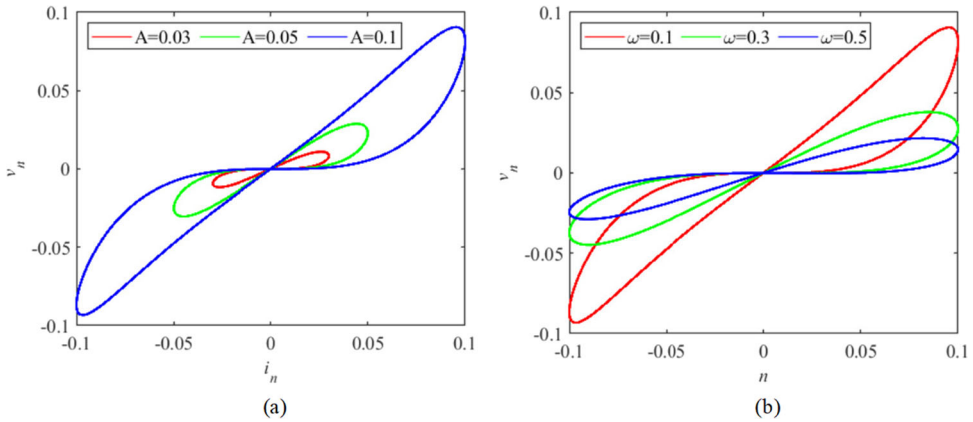


Fig. 1 Memory characteristics of memristor under different parameters of current source: **a** amplitude A is 0.03A, 0.05A, 0,1A; **b** frequency ω is 0.1ω, 0.3ω, 0.5ω

2.3 Discrete memristive hyperchaotic map

The chaotic range of the improved logistic map is wider than the original map, however it remains with the disadvantages of simple behavior and vulnerable chaotic intervals, which are not favorable for image encryption applications. The discrete memristive map can be viewed in the parameter range for more sophisticated chaotic behavior, but with a very narrow parameter range [36]. Combining with the above, the 2D-DMHM with abundant dynamical behavior can be achieved when coupling the improved Logistic map and the discrete memristor. The structure is shown in Fig. 2, where $G(u, x_n)$ denotes the improved Logistic map with control parameter u, the parameter k represents the coupling strength between the discrete memristor and the improved Logistic map. 2D-DMHM can be expressed as

$$\begin{cases} x_{n+1} = ux_n \sin(a(1 - x_n)) + k \sin(y_n)x_n \\ y_{n+1} = x_n + y_n \end{cases} \tag{4}$$

where $a = 0.1$, $k = 1.72$, and $u \in [-2.4, 4]$. As shown in Fig. 3(c), the 2D-DMHM is hyperchaotic in the range $u \in [-2, -0.1]$, which can be clearly found that the novel map has a large-scale hyperchaotic interval and exhibits excellent chaotic properties. The addition of the memristor raises the map from one to two dimensions, increasing the sequence randomness and the security of its application in image encryption. In comparison to the Logistic map and the improved Logistic map, the addition of the memristor also causes the 2D-DMHM to exhibit more complex hyperchaotic behavior.

2.4 Performance evaluation

The proposed 2D-DMHM can dramatically improve the chaotic performance of the improved Logistic map, which is partially validated more adequately by phase diagrams and Lyapunov exponential diagrams for its favorable chaotic performance, and compared with several existing 2D chaotic maps.

The phase diagram shows the motion of the system over time from the initial conditions, and the trajectory curve with chaotic behavior is never closed or repeated. The randomness of the output from the chaotic system can be inferred by analyzing the trajectory graph. The larger phase space occupied by the trajectory of the system signifies that the chaotic sequence has better randomness. Figure 4 compares the chaotic trajectory of 2D-DMHM with the 2D Logistic map and 2D-SLMM [37]. The control parameters are set to those that reflect the best chaotic performance of the corresponding maps. The comparison shows that the 2D-DMHM trajectory have a wider distribution space than the other maps, which means that the proposed map produces a greater range of outputs with better randomness.

Lyapunov exponent (LE) is a numerical characteristic to assess chaotic motions, which may quickly evaluate whether a map possesses chaotic qualities [38]. The hyperchaotic

Table 1 Chaotic map equations

Chaotic map	Equation	Initial condition
Gauss map	$x_{n+1} = e^{-\alpha x_n^2} + \beta$	$\alpha = 4.7, \beta = 0.6$
Tent map	$x_{n+1} = \begin{cases} \mu x_n, & for\ x_n < 1/2 \\ \mu(1 - x_n), & for\ x_n \geq 1/2 \end{cases}$	$\mu = 1.9, x_0 = 0.6$
Hénon map	$\begin{cases} x_{n+1} = 1 - ax_n^2 + y_n \\ y_{n+1} = bx_n \end{cases}$	$a = 1.4, b = 0.3$
Logistic map	$x_{n+1} = \mu x_n(1 - x_n)$	$\mu = 2$

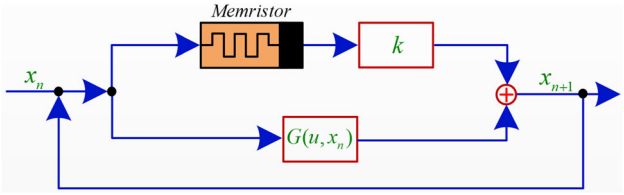


Fig. 2 Structure of the 2D-DMHM

map has at least two integer LEs while the ordinary chaotic map only has one positive LE. The more complex and unpredictably chaotic behavior of the hyperchaotic map can undoubtedly significantly increase the security of image encryption. Figure 5(a)-(c) show the variation of the two LEs values of the 2D Logistic map, 2D-SLMM, and 2D-DMHM with their control parameters respectively. The 2D Logistic map exhibits chaotic behavior at parameter $r \in [1.11, 1.15] \cup [1.18, 1.19]$, but no hyperchaotic behavior across the full parameter range, as shown in Fig. 4(a). The 2D-SLMM exhibits chaotic behavior at the control parameter $\alpha \in [0.84, 1]$ and hyperchaotic behavior at $\alpha \in [0.91, 1]$, as can be seen in Fig. 4(b). According to Fig. 4(c), 2D-DMHM exhibits chaotic behavior at $k \in [-2.4, 0.3]$ and hyperchaotic behavior at $k \in [-2.1, -0.1]$. The comparison demonstrates that the chaotic and hyperchaotic range of the proposed map is substantially larger than that of the other two maps, further showing the superior complexity, randomness, and unexpectedness of the 2D-DMHM.

2.5 NIST SP800-22 test

In order to show the feasibility and the applicability of the proposed 2D-DMHM in the design of the image encryption algorithm, the randomness of the hyperchaotic map export sequence are tested using the National Institute of Standards and Technology (NIST) SP800-22 standard. The P-value and the pass rate are utilized as the main metrics to determine if the 120 sets of binary chaotic sequence of length 10^6 produced by the proposed map are dispersed randomly. The explanation in [39] states that the sequence with high randomness should meet the following criteria when there are 120 groups of sequence and the level of significance $\alpha = 0.01$, the P-value must be within the range [0.01, 1] and the pass rate must be higher than 0.9628. Table 2 clearly depicts the results of the 15 sub-tests of the sequence, which shows that the sequence obtained by the proposed map fully fits the requirements of the random distribution.

3 Proposed encryption algorithm

Based on the previously discussed 2D-DMHM, a novel image encryption algorithm (DMHM-IEA) is suggested in this section, the structure is given in Fig. 6. Two separate chaotic matrices are produced for encryption after the initial values and parameters of the map are generated via key decompression. The algorithm generates an encrypted image that resembles noise out of the original image using the diffusion-permutation-diffusion structure. Effective position exchange is used to perform the permutation, whereby adjacent pixels are quickly moved to different positions, changing the position of each pixel twice on average. In the diffusion procedure, the image is diagonally diffused the rows and columns concurrently from two different directions, which can amplify tiny changes to affect the entire image.

3.1 Initial state generation

As a unique instrument to get the effective information of the encrypted image, The total key length of DMHM-IEA is

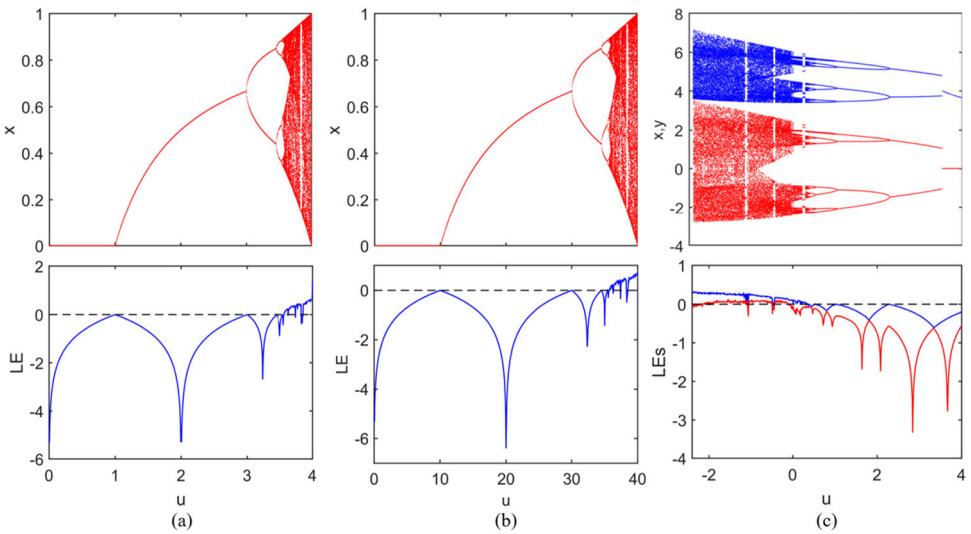


Fig. 3 The bifurcation and Lyapunov exponent diagrams of chaotic maps: **a** the Logistic map; **b** the improved Logistic map; **c** the 2D-DMHM

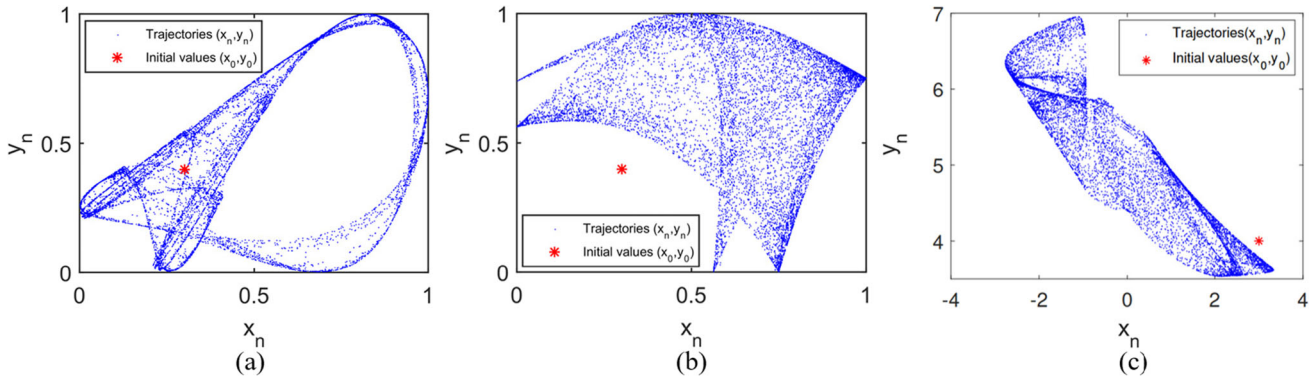


Fig. 4 Trajectory of different 2D chaotic maps: **a** the 2D Logistic map with parameter $r = 1.19$, $(x_0, y_0) = (0.3, 0.4)$; **b** the 2D-SLMM with parameter $\alpha = 1$, $(x_0, y_0) = (0.3, 0.4)$; **c** the 2D-DMHM with parameter $k = 1.72$, $u = -2$, $a = 0.1$, $(x_0, y_0) = (3, 4)$

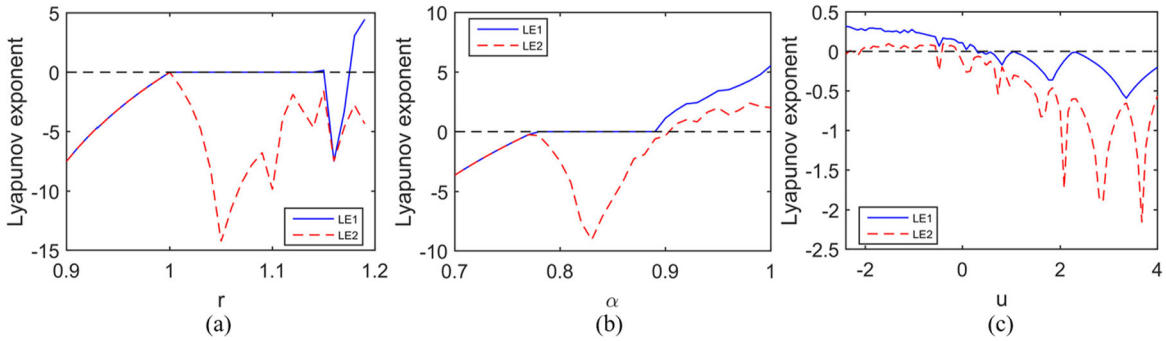


Fig. 5 The two LEs of different 2D chaotic maps: **a** the 2D Logistic map; **b** the 2D-SLMM; **c** the 2D-DMHM

Table 2 NIST SP800-22 Test Results of the hyperchaotic sequence

NO.	Sub-tests	P-value ≥ 0.01	Pass rate ≥ 0.9628	Result
01	Frequency	0.8603	0.9917	Pass
02	Block Frequency	0.5039	0.9917	Pass
03	Cumulative Sum (F)	0.7580	0.9917	Pass
	Cumulative Sum (R)	0.5948	0.9917	Pass
04	Runs	0.3144	0.9750	Pass
05	Longest Runs	0.5689	0.9750	Pass
06	Rank	0.5889	0.9917	Pass
07	FFT	0.5756	0.9750	Pass
08	Non-Overlapping Template*	0.1817	0.9667	Pass
09	Overlapping Template	0.6703	0.9917	Pass
10	Universal	0.6533	0.9833	Pass
11	Approximate Entropy	0.6480	0.9917	Pass
12	Random Excursions*	0.9332	0.9750	Pass
13	Random Excursions Variant*	0.9579	0.9750	Pass
14	Serial (P-value1)	0.6606	0.9833	Pass
	Serial (P-value2)	0.7148	0.9833	Pass
15	Linear Complexity	0.6468	1.0000	Pass

*The sub-tests Non-overlapping template, Random excursions and Random excursions variant contain 148, 8, 18 sub-tests

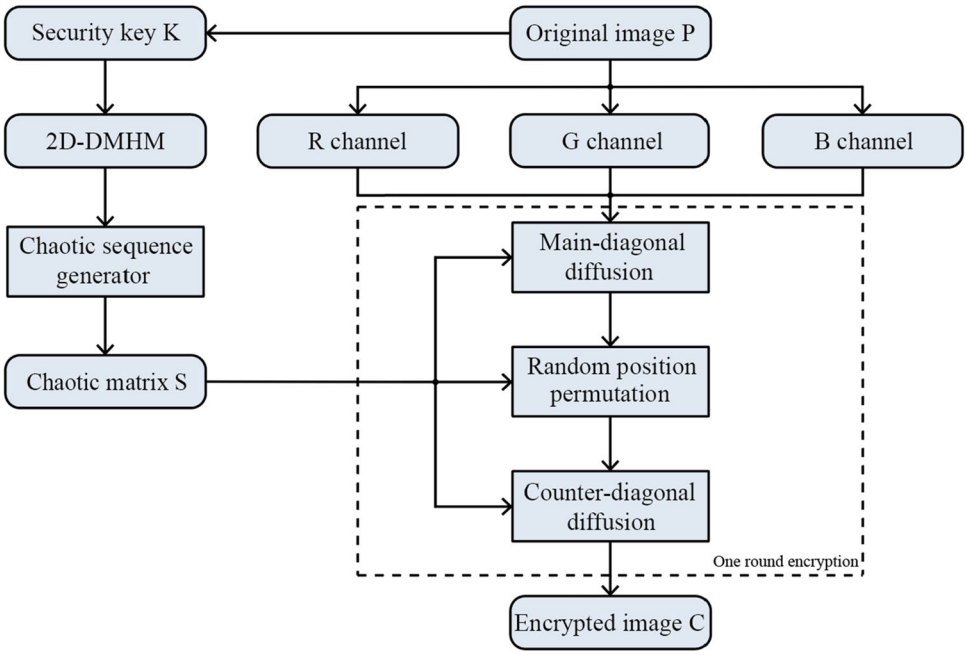


Fig. 6 Structure of the proposed DMHM-IEA

256 bits, and its structure is shown in Fig. 7. According to Algorithm 1, the initial state of hyperchaotic map is calculated from K, the initial values and parameters are produced by x, y, d_1 and d_2, r is the control perturbation variable, all of the variables are converted to floating point numbers using the IEEE754 format.

Algorithm 1 Generation of initial values and parameters.

Input: Security key K with length of 256 bits

1: Initial value $x \leftarrow (\sum_{i=1}^{52} K_i 2^{52-i})/2^{52}$;

2: Initial value $y \leftarrow (\sum_{i=53}^{104} K_i 2^{104-i})/2^{52}$;

3: Parameter $r \leftarrow (\sum_{i=105}^{156} K_i 2^{156-i})/2^{52}$;

4: **for** $j = 1$ to 2 **do**

5: Parameter $d_j \leftarrow (\sum_{i=106+50j}^{156+50j} K_i 2^{156+50j-i})/2^{50}$;

6: **end for**

7: $x_0 \leftarrow 0.5 + (x + r)\ mod\ 0.5$;

8: $y_0 \leftarrow 0.5 + (y + r)\ mod\ 0.5$;

9: $a_1 \leftarrow -2 + (d_1 + r)\ mod\ 0.5$;

10: $a_2 \leftarrow -2 + (d_2 + r)\ mod\ 0.5$;

Output: Initial conditions (x_0, y_0, a_1) and (y_0, x_0, a_2)

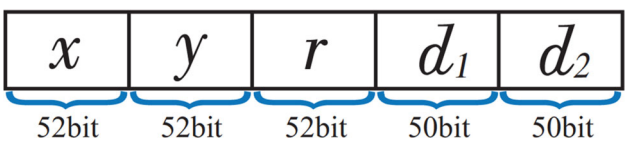


Fig. 7 Structure of the security key

3.2 Permutation process

The chaotic matrices produced by the proposed hyperchaotic map are sorted in various ways to generate four index matrices for the random position permutation process. The combination of the index matrices thus obtained can then create two different position exchange matrices, resulting in the realization of the random exchange of the pixel positions of the original image. The specific steps are as follows, and Algorithm 2 illustrates the permutation process.

- *Step* 1 : Assuming that the size of the original image is $m \times n$. A chaotic matrix $\mathbf{S}$ of size m×n is generated from the initial state by 2D-DMHM;
- *Step* 2 : Sort each column in the chaotic matrix $\mathbf{S}$ and obtain an index matrix $\mathbf{C}_{col}$, sort each row in the chaotic matrix $\mathbf{S}$ and obtain an index matrix $\mathbf{C}_{row}$;
- *Step* 3 : Sort the row t and the column t of chaos matrix $\mathbf{S}$, and obtain two index matrices $\mathbf{C}_{(t,:)}$ and $\mathbf{C}_{(:,t)}$ respectively. The t is generated from the initial parameters;
- *Step* 4 : Combine each column of $\mathbf{C}_{row}$ with $\mathbf{C}_{(:,t)}$ and obtain a position matrix $\mathbf{T}_1$, combine each row of $\mathbf{C}_{col}$ with $\mathbf{C}_{(t,:)}$ and obtain a position matrix $\mathbf{T}_2$;
- *Step* 5 : Exchange the positions corresponding to the two position matrices and obtain the permuted matrix $\mathbf{T}$.

In order to better understand the process of DMHM-IEA permutation, we have prepared an image of size 4×4 as an example. Figure 8 illustrates the position exchange matrices $\mathbf{T}_1$ and $\mathbf{T}_2$ obtained from the chaotic matrix $\mathbf{S}$. The rows and columns of the chaotic sequence $\mathbf{S}$ are first sorted to produce

Algorithm 2 The DMHM-IEA permutation.

Input: The original image $\mathbf{P}$ of size $m \times n$ and the security key $K = \{x, y, r, d_1, d_2\}$.

Generate chaotic matrix $\mathbf{S}$ using 2D-DMHM with initial state (x_0, y_0);
Sort each column of $\mathbf{S}$ to obtain the sorted matrix $\mathbf{C}_{col}$;
Sort each row of $\mathbf{S}$ to obtain the sorted matrix $\mathbf{C}_{row}$;
$t_1 \leftarrow (sum(\mathbf{S}_{floor(mod(a_1,n)+1)}) \times a_1) \ mod \ m + 1$;
$t_2 \leftarrow (sum(\mathbf{S}_{floor(mod(a_1,m)+1)}) \times a_1) \ mod \ n + 1$;
Sort the row t of $\mathbf{S}$ to obtain the sorted matrix $\mathbf{C}_{t_1}$;
Sort the column t of $\mathbf{S}$ to obtain the sorted matrix $\mathbf{C}_{t_2}$;
for $i = 1$ to m **do**
  for $j = 1$ to n **do**
    $tr = \mathbf{P}_{\mathbf{C}_{col(i,j)},\mathbf{C}_{t_1(j)}}$;
    $\mathbf{P}_{\mathbf{C}_{col(i,j)},\mathbf{C}_{t_1(j)}} = \mathbf{P}_{\mathbf{C}_{t_2(i)},\mathbf{C}_{row(i,j)}}$;
    $\mathbf{P}_{\mathbf{C}_{t_2(i)},\mathbf{C}_{row(i,j)}} = tr$;
  end for
end for
$\mathbf{T} = \mathbf{P}$

Output: The permutation result $\mathbf{T}$

the two index matrices $\mathbf{C}_{col}$ and $\mathbf{C}_{row}$. Second, by assuming that t equals 3, it is possible to generate the index matrices $\mathbf{C}_{(3,:)}$ and $\mathbf{C}_{(:,3)}$. The position exchange matrices $\mathbf{T}_1$ and $\mathbf{T}_2$ are then created by combining the index matrices $\mathbf{C}_{(3,:)}$ and $\mathbf{C}_{col}$, $\mathbf{C}_{(:,3)}$ and $\mathbf{C}_{row}$. In accordance with the position exchange matrices, the pixel positions of the original image are exchanged finally. Figure 9 represents the evolution of the image pixel positions before and after permutation.

3.3 Diffusion process

A superior encryption algorithm should have a significant diffusion property that can transform a tiny difference in the original image into a large difference in the encrypted image. The proposed diagonal diffusion provides simultaneous diffusion for rows and columns in two separate directions. These are the descriptions of the diagonal diffusion models in the two different directions, the main-diagonal diffusion model is shown in Fig. 10(a), and its mathematical formulation is (5), the counter-diagonal diffusion model is shown in Fig. 10(b), and its mathematical formulation is (6).

$$\mathbf{C}_{i,j} = \begin{cases} (\mathbf{T}_{i,j} + \mathbf{T}_{1,n} + \mathbf{T}_{m,1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i = 1,\ j = 1 \\ (\mathbf{T}_{i,j} + \mathbf{T}_{m,j} + \mathbf{C}_{1,j-1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i = 1,\ j \neq 1 \\ (\mathbf{T}_{i,j} + \mathbf{T}_{i,n} + \mathbf{C}_{i-1,1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i \neq 1,\ j = 1 \\ (\mathbf{T}_{i,j} + \mathbf{C}_{i-1,j} + \mathbf{C}_{i,j-1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i \neq 1,\ j \neq 1 \end{cases} \tag{5}$$

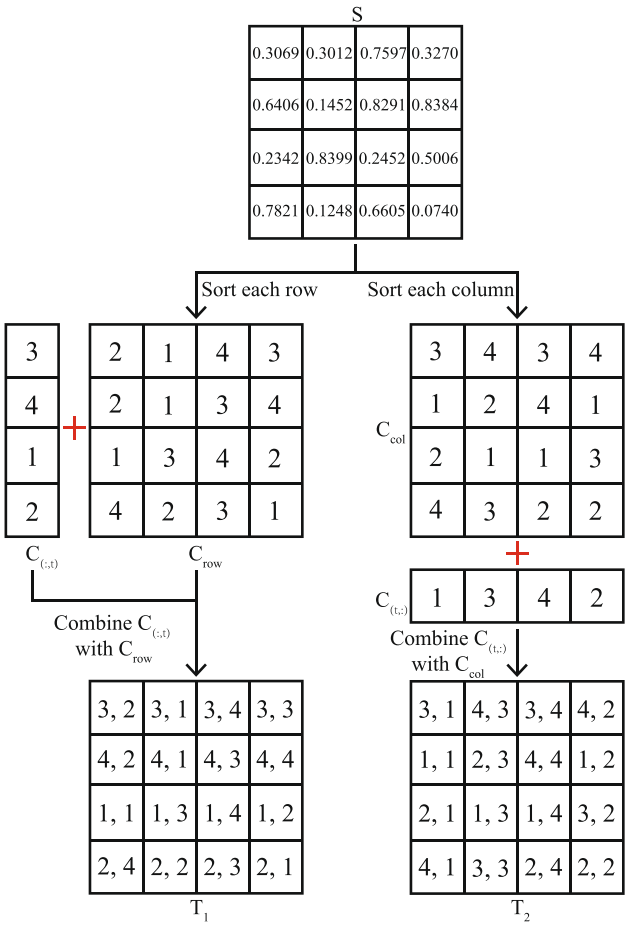


Fig. 8 The process of generating permutation matrices T_1 and T_2

$$\mathbf{C}_{i,j} = \begin{cases} (\mathbf{T}_{i,j} + \mathbf{T}_{1,n} + \mathbf{T}_{m,1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i = 1,\ j = n \\ (\mathbf{T}_{i,j} + \mathbf{T}_{m,j} + \mathbf{C}_{1,j+1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i = 1,\ j \neq n \\ (\mathbf{T}_{i,j} + \mathbf{T}_{i,n} + \mathbf{C}_{i-1,1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i \neq 1,\ j = n \\ (\mathbf{T}_{i,j} + \mathbf{C}_{i-1,j} + \mathbf{C}_{i,j+1} + \lfloor \mathbf{S}_{i,j} \times 2^{32} \rfloor) \bmod L, & \text{if } i \neq 1,\ j \neq n \end{cases} \tag{6}$$

where L is the maximum value of the image pixels, and the calculation $\lfloor x \rfloor$ is to obtain the maximum integer less than or

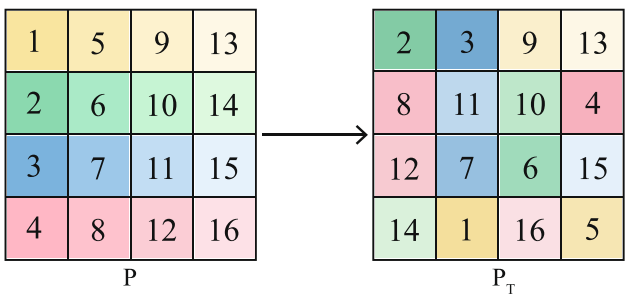


Fig. 9 The position of the image pixels before and after permutation

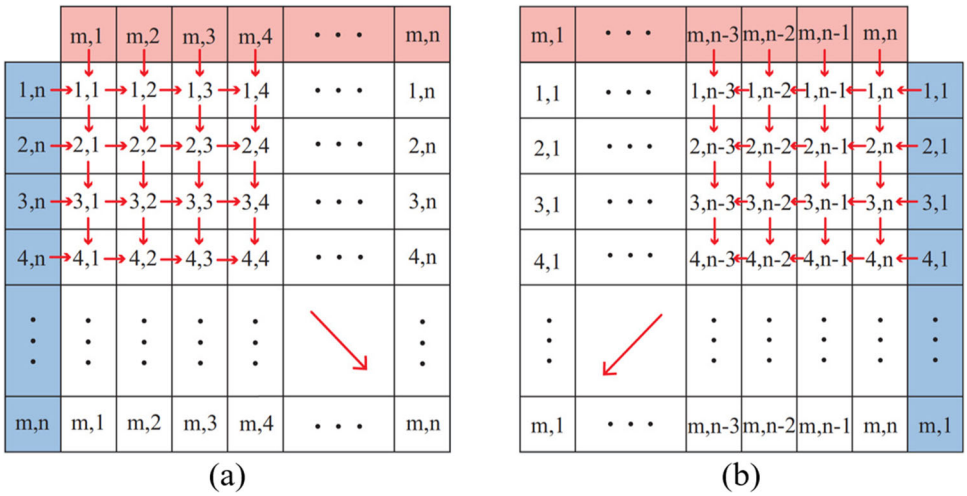


Fig. 10 The process of diffusion: **a** is main-diagonal diffusion; **b** is counter-diagonal diffusion

equal to x. We applied a diagonal diffusion on the original image in order to more clearly demonstrate the advantages of the diffusion, and Fig. 11 shows the local evolution as a result, which indicates that the diffused image has random grayscale values.

4 Simulation results and security analysis

This section simulates the proposed DMHM-IEA and analyzes the security performance of algorithm in great detail. Most of the test images in the experiment were selected from the USC-SIPI database. The experiments were conducted on a single computer in the following environment. Intel(R) Core(TM) i7-6700HQ CPU @ 2.60GHz, 8GB memory, Windows10 operating system.

4.1 Simulation results

Several types of original images should be able to be encrypted into unrecognizable encrypted images through a powerful and effective image encryption algorithm. The experimental simulation results for binary images, grayscale images, and color images are shown in Fig. 12. All-white and all-black images are two extreme examples of binary images, and because they can prevent the permutation from working, they are frequently used to crack the image encryption algorithms. The results show that DMHM-IEA has excellent encryption effect on different types of images.

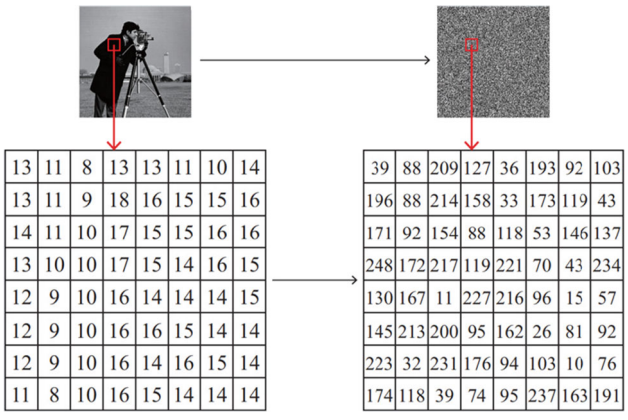


Fig. 11 An example of DMHM-IEA diffusion

4.2 Computational complexity analysis

The image encryption time can be used to evaluate the speed of the image encryption algorithm. The experiment encrypts a 256×256 grayscale image 100 times using DMHM-IEA and achieves an encryption time of 0.05178s, translating to an encryption speed of approximately 1.207Mb/s (Megabits per second). We use the same method as [40] to calculate the time complexity of DMHM-IEA. Assuming that the size of the original image is M×N, the complexity of generating chaotic sequences is O(2MN), the complexity of both main-diagonal diffusion and counter-diagonal diffusion is O(MN), and the complexity of random position permutation is O(2MN). Therefore, the time complexity of our proposed algorithm is O(6MN). Table 3 shows the time complexity of the different algorithms. In summary, the DMHM-IEA has high encryption efficiency.

4.3 Security analysis

An important aspect of evaluating the effectiveness of image encryption algorithm is security analysis. The security of DMHM-IEA is investigated in this section using histogram, pixel correlation, information entropy, ability to resist different attacks, key space and sensitivity, randomness, and robustness.

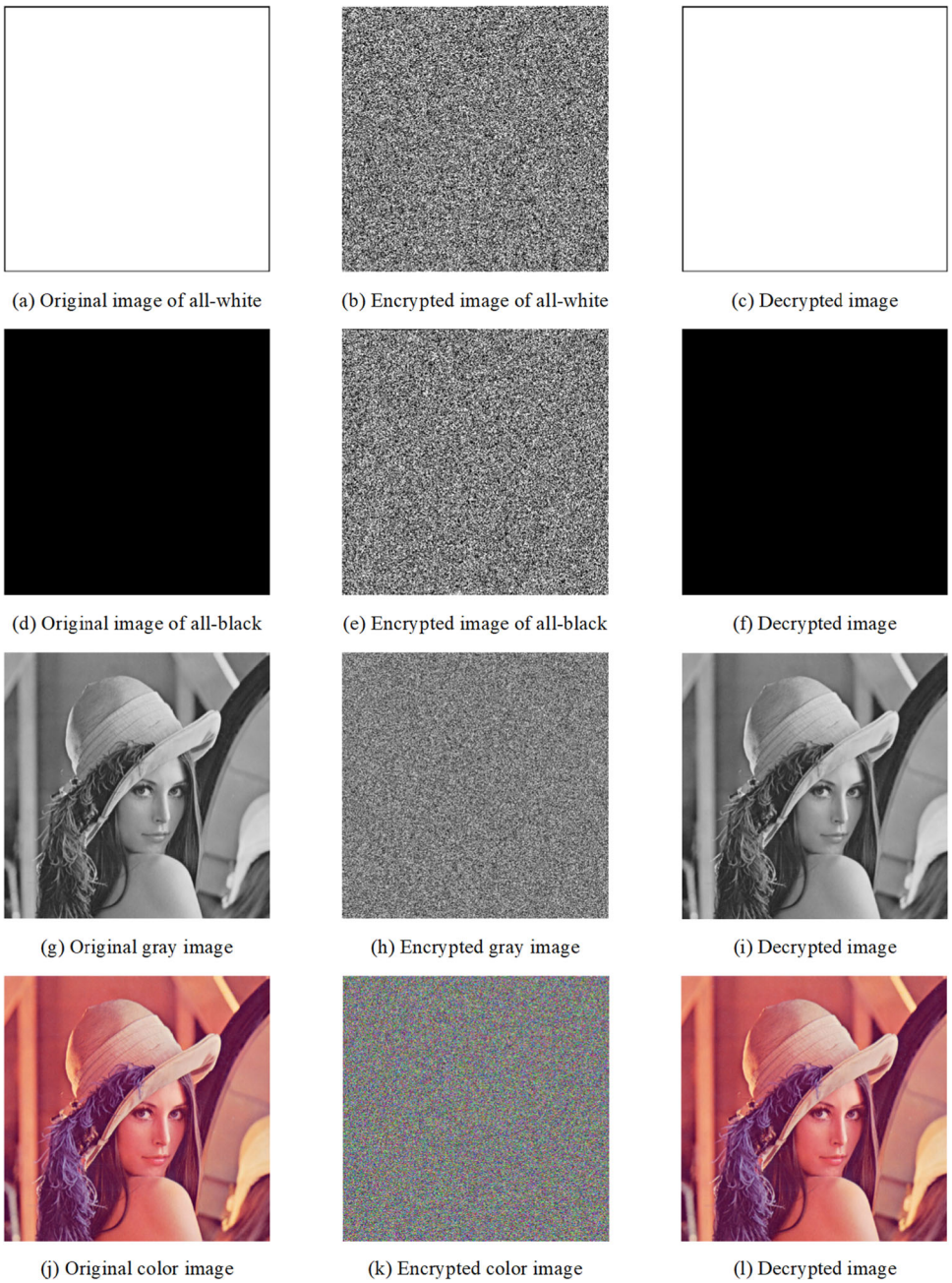


Fig. 12 Encryption simulations of DMHM-IEA

4.3.1 Histogram analysis

The distribution of pixel values in the image can be described using the histogram. A flat histogram means that the pixel values are evenly distributed and the readability of the image is very low. Figure 13 represents the histograms of all-white, all-black, gray and color original images and the corresponding encrypted images in Fig. 12. From the results, it can be seen that the encrypted images are not readable and their pixel values are uniformly distributed.

Table 3 Comparison of time complexity of different algorithms

Algorithms	Ref. [26]	Ref. [29]	Ref. [40]	**Proposed**
Time complexity	O(8MN)	O(8MN+3Mlog $\frac{MN}{3}$+3M+3N)	O(9MN)	O(6MN)

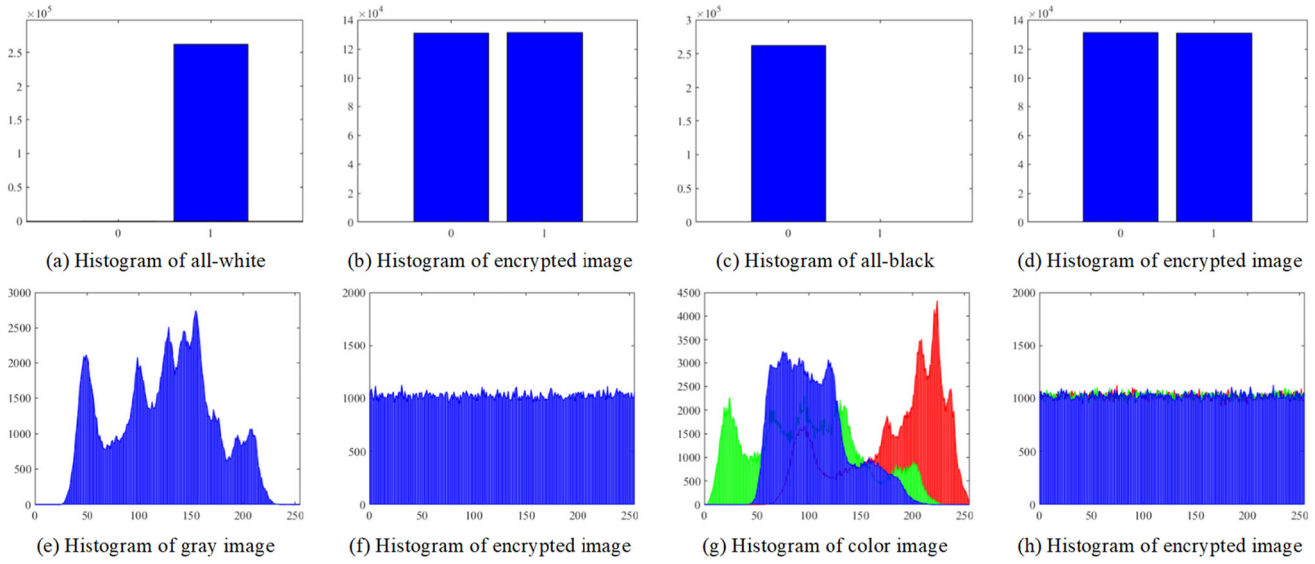


Fig. 13 Histograms of different original and encrypted images

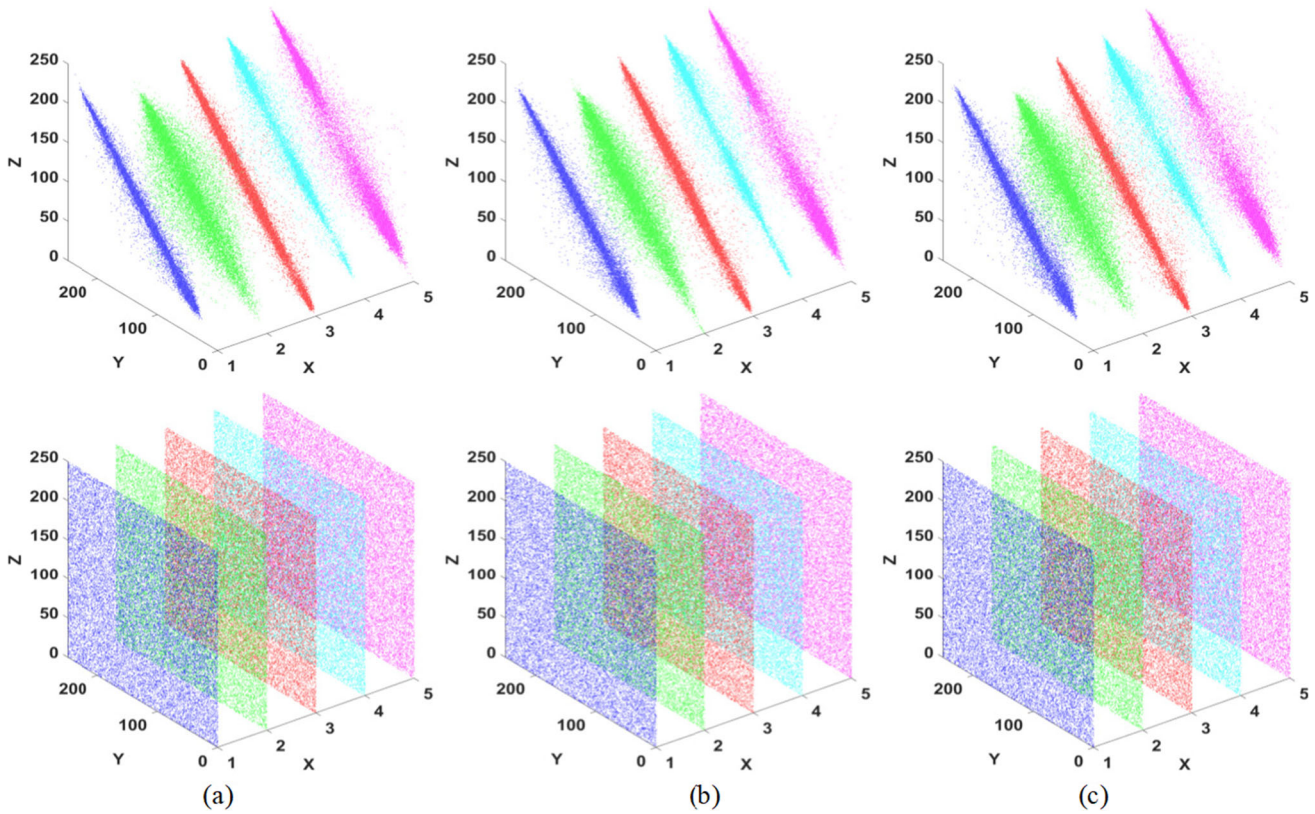


Fig. 14 Correlation of adjacent pixels in various directions for five different original and encrypted images: **a** horizontal; **b** vertical; **c** diagonal

Table 4 Adjacent pixel correlation of orginal image "Lena" in different image encryption algorithms

Schemes	Horizontal	Vertical	Diagonal
Lena	0.98890	0.98100	0.96800
Ref. [3]	-0.02540	0.00150	-0.00880
Ref. [20]	-0.00130	-0.00240	-0.00220
Ref. [21]	-0.00270	0.00230	0.00880
Proposed	-0.00230	0.00045	0.00530

4.3.2 Correlation analysis

The differences in values between adjacent pixels of the original image are small and highly correlated in horizontal, vertical and diagonal directions. The correlation between adjacent pixels of an image can be effectively broken by a good image encryption algorithm, making it impossible to decipher the encrypted image through statistical analysis. The correlation coefficient [41] between two adjacent pixels x and y can be expressed as

$$Cor = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2)(\frac{1}{N}\sum_{i=1}^{N}(y_i - \bar{y})^2)}} \tag{7}$$

where N is the number of sample points, $\bar{x}$ and $\bar{y}$ are the average of x_i and y_i, respectively. In order to show the correlation of adjacent pixels of images more obviously and intuitively, five different images are selected as test subjects, and Fig. 14 illustrates the correlation analysis of their adjacent pixels in different directions before and after encryption. Table 4 gives the results of correlation coefficients of encrypted images between different algorithms. It can be seen that the pixel distribution of the encrypted image is random and the correlation coefficient between adjacent pixels is very close to 0, which indicates that the proposed algorithm can resist statistical analysis attack.

Table 5 Information entropy for R,G,B channel of color orginal images and encrypted images

Image	R channel	G channel	B channel	**Mean**
4.1.01.tiff	7.9974	7.9974	7.9977	7.99750
4.1.02.tiff	7.9972	7.9974	7.9975	7.99737
4.1.03.tiff	7.9975	7.9976	7.9976	7.99757
4.1.04.tiff	7.9975	7.9973	7.9976	7.99747
4.1.05.tiff	7.9973	7.9978	7.9972	7.99743
4.1.06.tiff	7.9971	7.9972	7.9976	7.99730
4.1.07.tiff	7.9972	7.9979	7.9975	7.99753
4.2.01.tiff	7.9993	7.9994	7.9993	7.99933
4.2.02.tiff	7.9993	7.9993	7.9994	7.99933
4.2.03.tiff	7.9994	7.9993	7.9993	7.99933
4.2.04.tiff	7.9994	7.9994	7.9994	7.99940
4.2.05.tiff	7.9994	7.9994	7.9993	7.99937
4.2.06.tiff	7.9994	7.9993	7.9994	7.99937
4.2.07.tiff	7.9995	7.9993	7.9993	7.99937

4.3.3 Information entropy analysis

Another crucial metric to assess the distribution of image gray values and to gauge the randomness of image information is information entropy [42], which can be expressed as

$$H(x) = -\sum_{i=0}^{2^L-1} p(x_i)\log_2 p(x_i) \tag{8}$$

where $p(x_i)$ denotes the frequency of information occurrence, and for the image with gray level of 8, $L = 8$. Information entropy has a theoretical value of 8, so the closer a result is to 8 the more random the gray value distribution of the image is, and the less probable it is that information will leak. The experimental testing results shown in Table 5 show that the proposed algorithm has high randomness and low information leakage probability, which is sufficient to withstand statistical analysis attack.

Table 6 NPCR and UACI of different images encrypted by DMHM-IEA

File Name	NPCR	UACI	File Name	NPCR	UACI
4.1.01.tiff	99.6140	33.4598	**5.1.14.tiff**	99.6093	33.4645
4.1.02.tiff	99.6094	33.4737	**5.2.08.tiff**	99.6051	33.4624
4.1.03.tiff	99.6017	33.4709	**5.2.09.tiff**	99.6089	33.4650
4.1.04.tiff	99.6124	33.4694	**5.2.10.tiff**	99.6074	33.4677
4.1.05.tiff	99.6078	33.4679	**5.3.01.tiff**	99.6128	33.4635
4.1.06.tiff	99.6074	33.4666	**5.3.02.tiff**	99.6097	33.4637
4.1.07.tiff	99.6135	33.4673	**7.1.01.tiff**	99.6120	33.4622
4.2.01.tiff	99.6120	33.4661	**7.1.02.tiff**	99.6109	33.4622
4.2.02.tiff	99.6029	33.4663	**7.1.03.tiff**	99.6131	33.4647
4.2.03.tiff	99.6075	33.4687	**7.1.04.tiff**	99.6078	33.4619
4.2.04.tiff	99.6070	33.4635	**7.1.05.tiff**	99.6143	33.4623
4.2.05.tiff	99.6048	33.4611	**7.1.06.tiff**	99.6121	33.4635
4.2.06.tiff	99.6082	33.4620	**7.1.07.tiff**	99.6086	33.4648
4.2.07.tiff	99.6105	33.4633	**7.1.08.tiff**	99.6055	33.4613
5.1.09.tiff	99.6078	33.4636	**7.1.09.tiff**	99.6055	33.4675
5.1.10.tiff	99.6139	33.4639	**7.1.10.tiff**	99.6070	33.4633
5.1.11.tiff	99.6071	33.4664	**7.2.01.tiff**	99.6068	33.4652
5.1.12.tiff	99.6032	33.4663			
5.1.13.tiff	99.6093	33.4656	**Mean**	99.6088	33.4649

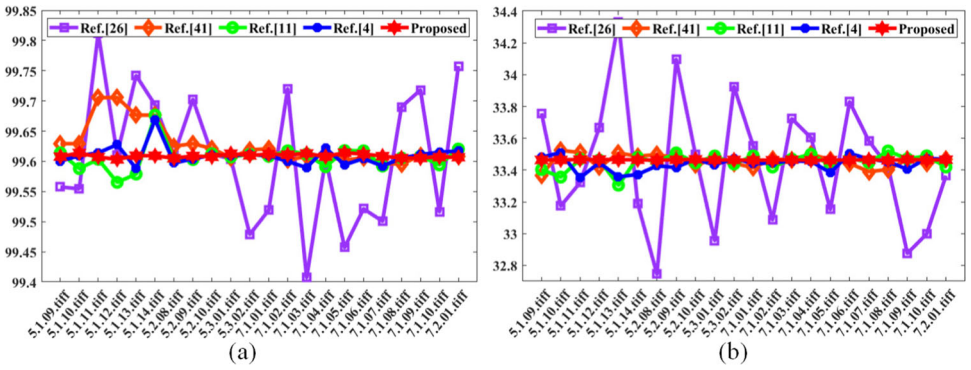


Fig. 15 Comparison of different image encryption algrithms: **a** is the NPCR comparison results; **b** is the UACI comparison results

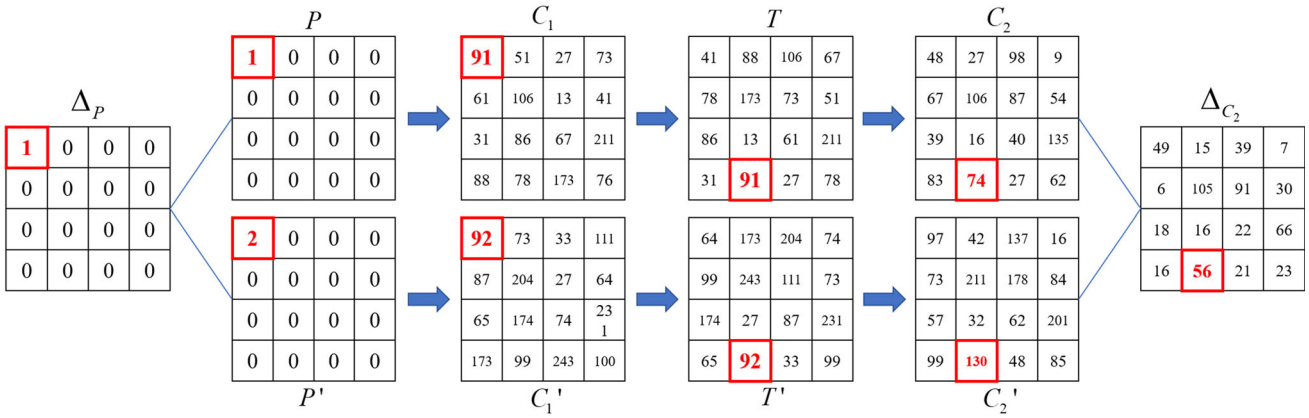


Fig. 16 The process of revealing the DMHM-IEA with a differential attack

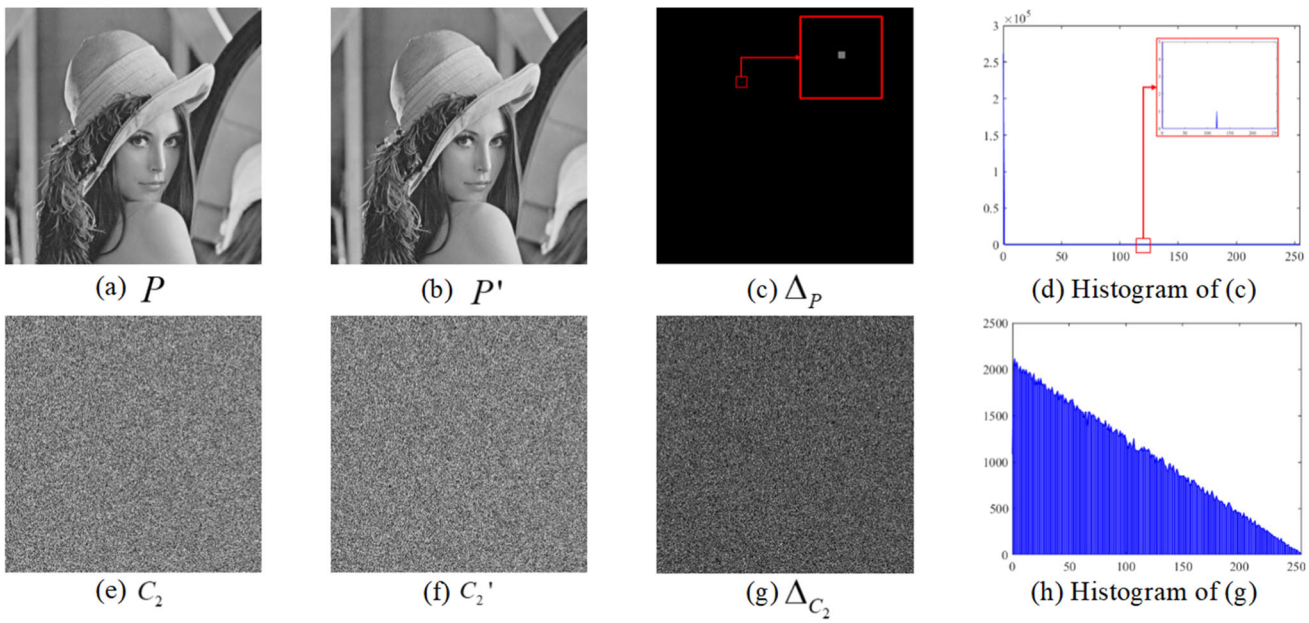


Fig. 17 Demonstration of resistance to chosen-plaintext attack ability

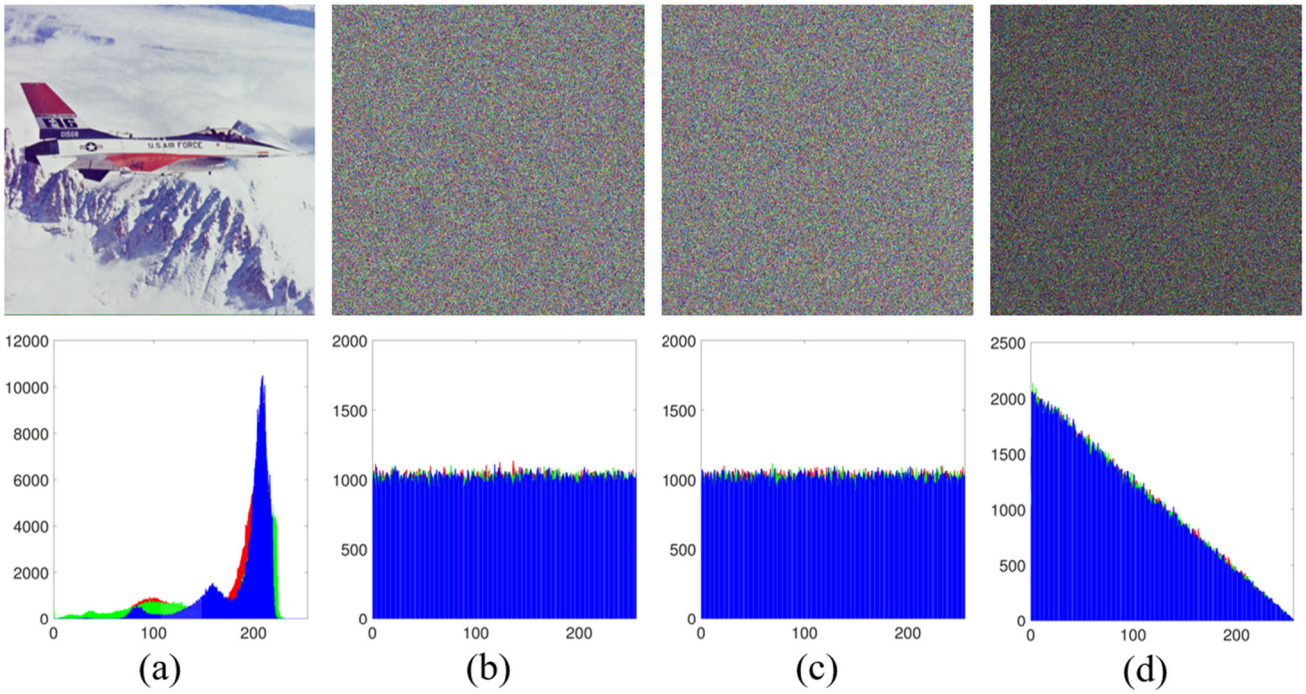


Fig. 18 Key sensitivity analysis of encryption processes: **a** original image **P**; **b** encrypted result $\mathbf{C}_1$ using K_1; **c** encrypted result $\mathbf{C}_2$ using K_2; **d** the difference between $\mathbf{C}_1$ and $\mathbf{C}_2$

4.3.4 Differential attack analysis

The number of pixel change rate (NPCR) and unified averaged changed intensity (UACI) can be used to evaluate whether the image encryption algorithm has excellent permutation and diffusion properties, which are commonly used to evaluate the performance of image encryption algorithm in resisting differential attacks. In mathematical terms, the NPCR and UACI [43] between two images $\mathbf{C}_1$ and $\mathbf{C}_2$ can be defined as

$$\mathrm{NPCR}(\mathbf{C}_1, \mathbf{C}_2) = \sum_{i=1}^{\mathrm{m}} \sum_{j=1}^{\mathrm{n}} \frac{\mathbf{D}(i,j)}{\mathrm{m} \times \mathrm{n}} \times 100\% \tag{9}$$

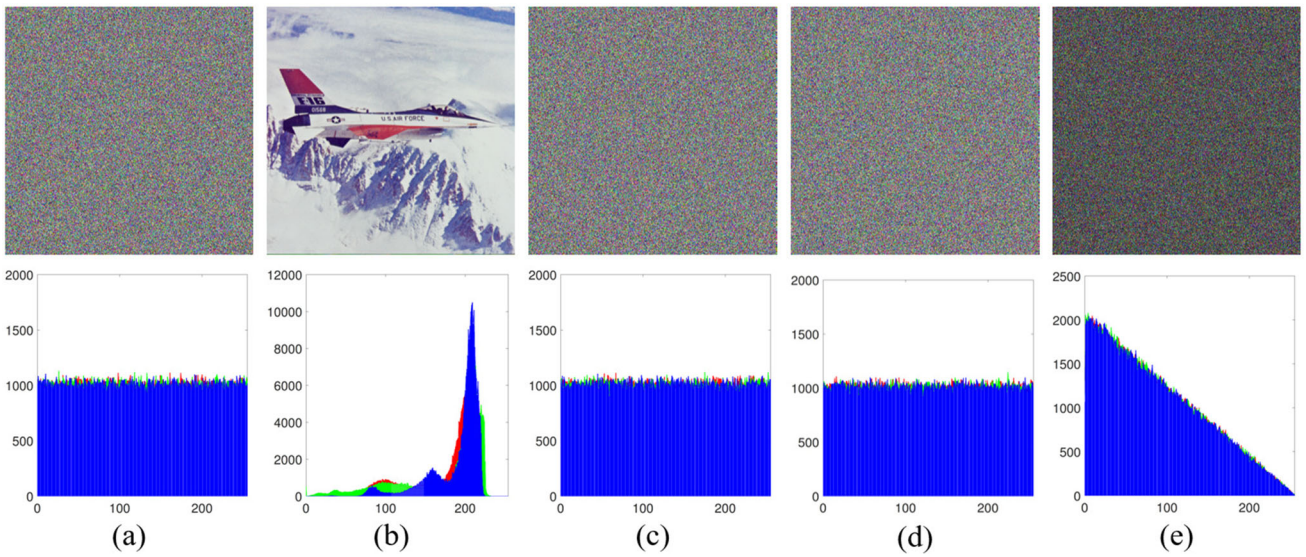


Fig. 19 Key sensitivity analysis of decryption processes: **a** encrypted image $\mathbf{C}_1$; **b** decrypted result **D** using K_1; **c** decrypted result $\mathbf{D}_1$ using K_2; **d** decrypted result $\mathbf{D}_2$ using K_3; **e** the difference between $\mathbf{D}_1$ and $\mathbf{D}_2$

Table 7 NIST SP800-22 Test Results of the encrypted images

NO.	Sub-tests	P-value ≥ 0.01	Pass rate ≥ 0.9628	Result
01	Frequency	0.4915	0.9833	Pass
02	Block Frequency	0.6661	1.0000	Pass
03	Cumulative Sum (F)	0.6074	1.0000	Pass
	Cumulative Sum (R)	0.3676	1.0000	Pass
04	Runs	0.9996	0.9917	Pass
05	Longest Runs	0.7051	0.9833	Pass
06	Rank	0.7371	1.0000	Pass
07	FFT	0.7136	1.0000	Pass
08	Non-Overlapping Template*	0.4817	0.9667	Pass
09	Overlapping Template	0.9817	0.9750	Pass
10	Universal	0.9922	0.9833	Pass
11	Approximate Entropy	0.6215	1.0000	Pass
12	Random Excursions*	0.3266	0.9655	Pass
13	Random Excursions Variant*	0.8010	0.9655	Pass
14	Serial (P-value1)	0.7224	0.9917	Pass
	Serial (P-value2)	0.9601	0.9917	Pass
15	Linear Complexity	0.6064	0.9833	Pass

*The sub-tests Non-overlapping template, Random excursions and Random excursions variant contain 148, 8, 18 sub-tests

$$\mathbf{D}(i,j) = \begin{cases} 0, & \text{if } \mathbf{C}_1(i,j) = \mathbf{C}_2(i,j) \\ 1, & \text{if } \mathbf{C}_1(i,j) \neq \mathbf{C}_2(i,j) \end{cases} \tag{10}$$

$$\mathrm{UACI}(\mathbf{C}_1, \mathbf{C}_2) = \sum_{i=1}^{\mathrm{m}} \sum_{j=1}^{\mathrm{n}} \frac{|\mathbf{C}_1(i,j) - \mathbf{C}_2(i,j)|}{\mathrm{L} \times (\mathrm{m} \times \mathrm{n})} \times 100\% \tag{11}$$

where where m and n are the number of rows and columns of the image, i and j are the row and column values of the current pixel, $\mathbf{D}$ is an array of the same size as images $\mathbf{C}_1$ and $\mathbf{C}_2$, and L is the maximum pixel value of the image. In this experiment, 14 color images along with 22 grayscale images are used for the test. By changing any one pixel point of each original image $\mathbf{P}_1$, another image $\mathbf{P}_2$ can be obtained, and then encrypting $\mathbf{P}_1$ and $\mathbf{P}_2$ with the same key by using DMHM-IEA, two encrypted images $\mathbf{C}_1$ and $\mathbf{C}_2$ can be obtained respectively, and the NPCR and UACI values of encrypted images $\mathbf{C}_1$ and $\mathbf{C}_2$ can be calculated, with all data recorded in Table 6. The ideal values of NPCR and UACI are 99.6094% and 33.4635%, and the experimental data show that the results of the DMHM-IEA are very close to the theoretical values. Figure 15(a) and (b) express the NPCR and UACI values of the different image encryption algorithms using the same original image to encrypt respectively. Compared with the other algorithms, the numbers obtained by the proposed DMHM-IEA are more stable and have a smaller range of fluctuation. The comprehensive results can be gained that DMHM-IEA has high resistance to differential attacks which can effectively ensure image information security.

4.3.5 Potential attacks analysis

The ability of the image encryption algorithm to withstand potential attacks, such as ciphertext-only attack, known-plaintext attacks, chosen-plaintext attack, and chosen-ciphertext attack, is crucial, especially when the attacker is aware of all the designs and operations aside from the key. As the strongest of the potential attacks, an image encryption algorithm can resist the chosen-plaintext attack, which means it can resist other potential attacks. The differential attack analysis is a chosen-plaintext attack [44], and the proposed DMHM-IEA for a round of differential attack analysis in this paper can be summarized as follows.

Step 1: Assume two original images P and P' with sizes m×n meet the requirements.

$$\begin{cases} P(i^*, j^*) = a \\ P'(i^*, j^*) = b \\ P(i,j) = P'(i,j) = c \end{cases} \tag{12}$$

where $a, b, c \in [0, 255]$, $(i, j) \in \{(1,1), (1,2), ..., (M, N)\}$, (i^*, j^*) is one of the set of (i, j), and the initial value of (i^*, j^*) is (1, 1), the difference between P and P' can be expressed as Δ_P

Step 2: The main-diagonal diffusion images C_1 and C_1', the random position permutation images T and T', and the counter-diagonal diffusion images C_2 and C_2'

are obtained respectively by encryption operation of DMHM-IEA for P and P' using unknown keys.
Step 3: Obtain the value of Δ_{C_2} using C_2 and C_2', and $\Delta_{C_2} = C_2'(i, j) - C_2(i, j)$
Step 4: Repeat the step 1∼3 for $(i^*, j^*) = (1, 2), (1, 3), \ldots, (m, n)$.

Figure 16 gives the observations of the two encrypted images obtained using DMHM-IEA encryption when $(i^*, j^*) = (1, 1)$ is the only non-zero element. It can be seen that even if the original image is filtered, the uniqueness of the proposed algorithm structure prevents the existence of security flaws, making it impossible to determine a valid location or piece of information.

The aforementioned procedure is simulated in Fig. 17 to show how well DMHM-IEA can withstand such attacks. The results show a noticeable difference between the two encrypted images, suggesting that the proposed algorithm has a strong defense against potential attacks, including some plaintext attacks. This is also illustrated by the encryption of the all-white and all-black images in Fig. 12.

4.3.6 Key space and sensitivity analysis

The key space of the image encryption algorithm based on chaotic map must reach 2^{100} to be sufficient to resist the key attack of the exhaustive method [45]. If the key space is too small, it will not be able to withstand the attack using the exhaustive method and will be simple to crack. The length of the encryption process and the efficiency of the encryption algorithm will both suffer from an excessively large key space. Considering the security and efficiency of the image encryption algorithm, DMHM-IEA sets the key space to 2^{256}.

The sensitivity of the key ensures that minor changes can lead to significant differences in the encrypted and decrypted images. It may happen that the incorrect key is used to obtain the correct image information if the key is not sensitive enough, in which case the actual key space is smaller than the



Fig. 20 Analysis of the decryption effect of different noises: **a** 1% Salt & Pepper noise; **b** 3% Salt & Pepper noise; **c** 5% Salt & Pepper noise; **d** 1% Speckle noise; **e** 0.3% Gaussian noise; **f** Poisson noise

theoretical value. Keys K_1, K_2 and K_3 are identical except for a single arbitrary bit. Figure 18 shows how sensitive the keys are during the encryption process. By using the keys K_1 and K_2, the original image is encrypted to create $\mathbf{C}_1$ and $\mathbf{C}_2$, which have NPCR of 99.6048% and UACI of 33.4652%. In Fig. 19, the decryption of the image encrypted by key K_1 using K_2 and K_3 results in the decrypted images $\mathbf{D}_1$ and $\mathbf{D}_2$, which have NPCR of 99.6099% and UACI of 33.4605%. This illustrates the key sensitivity in the decryption process. The aforementioned findings demonstrate that the DMHM-IEA has extremely high key sensitivity because they demonstrate how completely different the images produced by it using nuanced keys for encryption and decryption are.

4.3.7 Randomness analysis

The random distribution of pixels in an encrypted image serves as a gauge for how effectively the image encryption algorithm can protect sensitive information. The experiment uses DMHM-IEA to encrypt 120 images of size 256×256, and the first 10^6 binary numbers of each image are selected as a set of test sequences for the 15 randomness sub-tests of NIST SP800-22. Sequences with randomness should have a P-value greater than 0.01 at the significance level of $\alpha = 0.01$, and the pass rate should be higher than 96.28%. The test results are displayed in Table 7, which unmistakably demonstrates that DMHM-IEA passed all randomness tests. This shows that the encrypted image of the proposed algorithm has strong randomness and further indicates the high security performance of DMHM-IEA.

4.3.8 Robustness analysis

During network transmission, digital images are vulnerable to a wide range of signal disturbances and data loss to varying degrees. Therefore, an excellent image encryption algorithm should be able to minimize the impact of noise and data loss on the acquired image information. The robustness of DMHM-IEA to noise is first analyzed, and Fig. 20 illustrates the degree of recovery of decrypted image data under the influence of 1%, 3% and 5% Salt & Pepper, 1% Speckle noise, 0.3% Gaussian noise and Poisson noise for encrypted images. The robustness of the proposed algorithm against data loss is analyzed next, and Fig. 21 shows the recovery

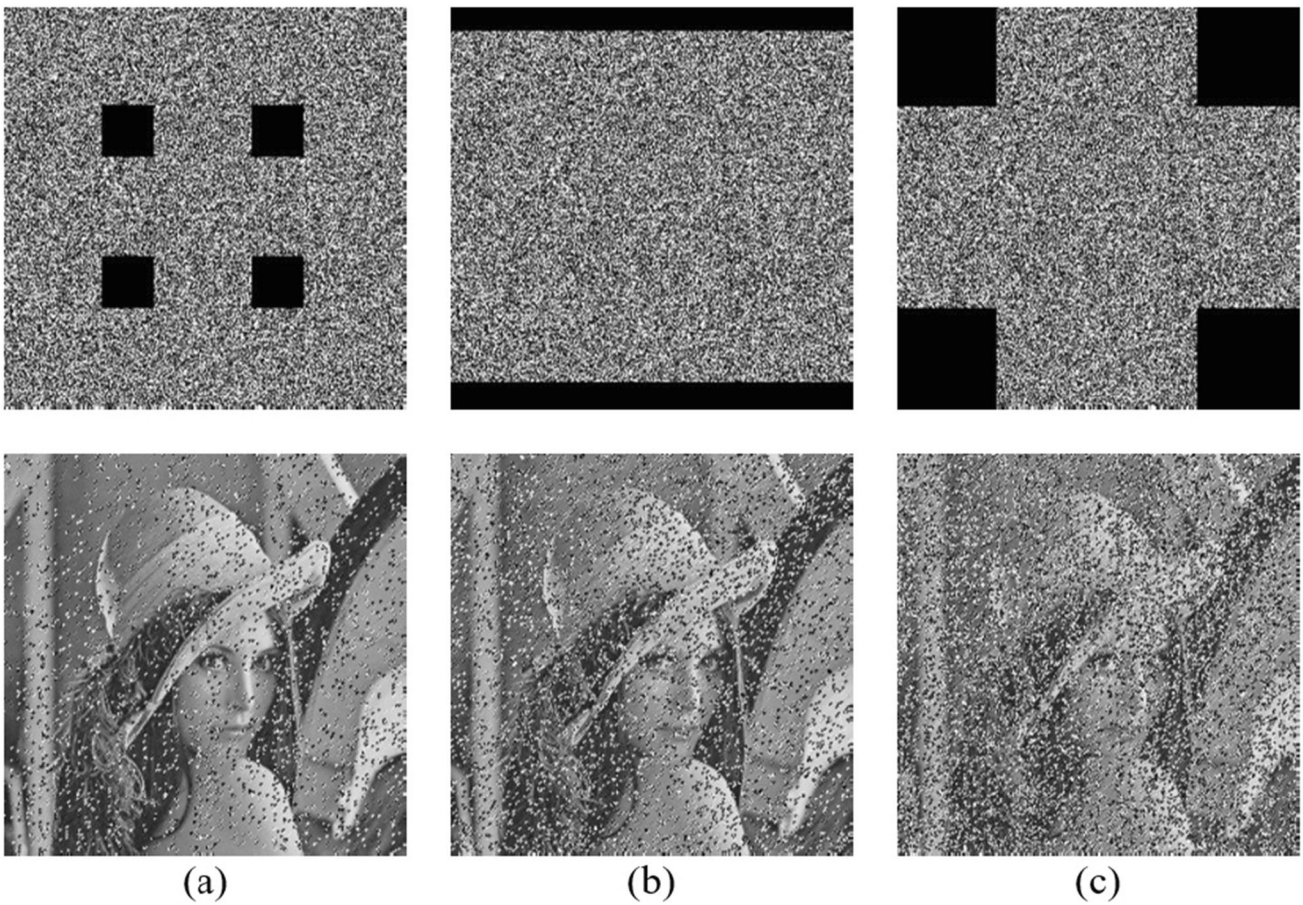


Fig. 21 Data loss analysis of encrypted images and their decrypted images: **a** 6.25% data loss; **b** 12.5% data loss; **c** 25% data loss

of the decrypted image for 6.25%, 12.5% and 25% data loss of the encrypted image. According to the decryption results, DMHM-IEA has a strong robustness and practical application value because it can decrypt and obtain crucial information despite extremely strong noise interference and significant data loss.

5 Conclusions

In this paper, a new two-dimensional discrete memristive hyperchaotic map with simple algebraic structure, complex dynamical behavior, high computational efficiency and high randomness is obtained by incorporating the discrete memristor into the improved Logistic map, and it is extremely suitable for image encryption applications. Therefore, based on the proposed map, we design a novel image encryption algorithm that employs the diffusion-displacement-diffusion structure, which includes two kinds of diagonal diffusion in different directions and a random position permutation. It can be seen through simulation experiments and comparative analysis that the algorithm is quick, highly sensitive to keys, secure, and reliable. Through simulation experiments and comparative analysis, it can be observed that the algorithm is characterized by high speed, strong key sensitivity, safety and reliability, which can effectively resist potential attacks, such as statistical analysis attack and chosen-plaintext attack, and has strong robustness against noise interference and data loss during transmission. Designing image encryption algorithms based on the chaotic map of coupled memristor is an interesting and meaningful research topic. We will further study the effects of different memristors on the chaotic map, optimize the structure of the algorithm, and design secure and efficient image encryption algorithm.

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Data Availability All data generated or analysed during this study are included in this published article.

Declarations

Conflict of Interests The authors declare that they have no known competing financial interestsor personal relationships that could have appeared to influence the work reported in this article.

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